

Vel Tech Group of Institutions

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2028_NeoColab_Design and Analysis of AlgorithmsS1L3

VTU_DAA_Week 8_MCQ

Attempt : 1

Total Mark : 15

Marks Obtained : 11.5

Section 1 : MCQ

1. Which type of graphs can be topologically sorted?

Answer

Directed graphs

Status : Correct

Marks : 1/1

2. Topological sort can be used to detect which of the following?

Answer

Cycles in a directed graph

Status : Correct

Marks : 1/1

3. In topological sort, vertices are visited in which order?

Answer

In topological order

Status : Correct

Marks : 1/1

4. Which of the following data structures can be used to implement topological sort?

Answer

All of the listed options

Status : Wrong

Marks : 0/1

5. Which of the following algorithms can be used to perform topological sort?

Answer

Depth First Search (DFS)

Status : Partially correct

Marks : 0.5/1

6. Topological sort is equivalent to which of the traversals in trees?

Answer

Post-order traversal

Status : Wrong

Marks : 0/1

7. When is the topological sort of a graph unique?

Answer

When there exists a hamiltonian path in the graph

Status : Correct

Marks : 1/1

8. In Dijkstra's algorithm, if two vertices u and v have the same tentative distance from the source, which one is chosen first?

Answer

It can be arbitrary

Status : Correct

Marks : 1/1

9. In a graph with non-negative weights, why is Dijkstra's algorithm guaranteed to find the shortest path?

Answer

Because once a vertex's distance is finalized, it cannot be improved

Status : Correct

Marks : 1/1

10. How can Dijkstra's algorithm be adapted to return not only distances but also the actual paths?

Answer

Maintain a parent array for each vertex

Status : Correct

Marks : 1/1

11. In Dijkstra's algorithm, what is the purpose of the relaxation step when processing an edge (u, v) ?

Answer

To update the distance to vertex v if a shorter path through u is found

Status : Correct

Marks : 1/1

12. In a graph with multiple edges between the same pair of vertices, how should Dijkstra's algorithm handle them?

Answer

Consider only the edge with the minimum weight

Status : Correct

Marks : 1/1

13. If a graph has an edge weight of zero, will Dijkstra's algorithm still work correctly?

Answer

Yes, always

Status : Correct

Marks : 1/1

14. Which property of Dijkstra's algorithm guarantees that the shortest path to a vertex is correct when the vertex is added to the finalized set?

Answer

Greedy choice property

Status : Wrong

Marks : 0/1

15. Why can Dijkstra's algorithm fail on a graph with a negative weight edge even if there are no negative cycles?

Answer

Because a previously finalized vertex may get a shorter path later

Status : Correct

Marks : 1/1